ICSI499 Capstone Project Report

BIAS In NLP

Project Team

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Abstract

Source texts and encyclopedias should be as objective as possible, because bias could render their information inaccurate. Nonetheless, biases permeate every part of human communication, so it is absolutely necessary to understand their nature and to be able to detect their presence. Techniques for natural language processing are critical in our daily lives. In spite of these methods having been shown to be successful in a variety of ways, they may inadvertently or consciously reproduce existing biases in the data, such as gender bias. The many problems that have surfaced since the popular vector representation of text data was applied to tasks like machine learning and natural language processing include instances have to do with its vulnerability to becoming embedded. In our project, we will using word embedding and cosine similarity method to determine the bias and divide them into five domains. These findings may result in an increase in the accuracy of natural language processing systems. Source texts and encyclopedias should be as objective as possible, because bias could render their information inaccurate. Nonetheless, biases permeate every part of human communication, so it is absolutely necessary to understand their nature and to be able to detect their presence. Techniques for natural language processing are critical in our daily lives. In spite of these methods having been shown to be successful in a variety of ways, they may inadvertently or consciously reproduce existing biases in the data, such as gender bias. The many problems that have surfaced since the popular vector representation of text data was applied to tasks like machine learning and natural language processing include instances have to do with its vulnerability to becoming embedded. In our project, we will use word embedding and cosine similarity methods to determine the bias and divide them into five domains. These findings may result in an increase in the accuracy of natural language processing systems.

Contents

1 Introduction **1**

2 Background and Related Work **3**

3 Proposed System/Application **5**

3.1 An Overview 5

3.2 System Requirements 6

3.3 Technical Design 7

3.4 System Implementation 9

3.5 Use of Computer Science Theory and Software Development Fundamentals . . 10

4 Experimental Design and Testing **11**

5 Ethics and Legal Practices **16**

6 Effort Sharing **17**

7 Conclusion and Future Work **18**

Bibliography **20**

1. Introduction

Natural Language Processing refers to the task of using a machine learning algorithm to analyze and interpret patterns in text-based datasets. Because NLP is based on machine learning models, although the machine does not have emotions, the machine learning model will also have some bias in the processing of language. For this project, we try to determine how biased the chosen dataset is by vectorizing the input data through the model and perform calculations. Discovering and effectively dealing with deviations in the text is a very important part of NLP. Nowadays, through machine learning, many better models for natural language processing have been developed, but there are still deviations. Although machines do not have emotions, they train and organize data through machine learning. In the end, there will still be unobvious bias. These deviations will greatly affect the machine's next operation. For example, a recent study discovered that algorithms used to predict repeat offenders have an unintentional racial bias [1]. Because language is one of the most complicated things invented by mankind. The language contains many elements and human emotions, so it is very difficult to judge and deal with language deviations. People use dialects and a variety of word choices in social media, based on their characteristic, which use a variety of terms to describe the same subject or situation [2]. Before starting the project, we found a way to judge the bias of the information vector. This method is using 0 and 1 to identify the presence or absence of a word in a sentence. At this time, each word can be represented by a vector of length V (the number of vocabularies in the corpus). On this basis, the vector of each word in a sentence is added to obtain the bag-of-words model. This method has problems because the bag-of-words model is that it is impossible to know whether there is a correlation between different words, and the meaning of words is often related to the context. Our core ideas are that in the first step we need to find a very suitable data set, and secondly, we need to use the existing function to vectorize our data set, and then process our vectorized data set accordingly to get what we want.

The key characteristics of the proposed solution are We are committed to finding the right data set for our model and use different methods for the vector words of the text data set. And we have selected 5 different ranges, such as finance. We conducted a bias study on the words in these 5 intervals. Our solution is more targeted and efficient than other solutions. Because we are dealing with the deviation of the related words in the 5 intervals we selected so that we can better find the existence of the deviation in these 5 intervals. Our model is not a

treatment of general bias, so it is more efficient in the corresponding field. After this project, we have a new understanding of the bias that exists in NLP. The contribution of our project is that we determined a lot of bias in the vocabulary of the five selected intervals. The structure of our project is measuring bias using math function for data that has been vectorized, then measure bias using math function for data that has been vectorized. The next step is finding some existing experimental methods for bias through the calculation of mathematical models and training of existing data sets. Finally, implementing our own experimental method and perform the above operations to achieve our goal of determining the bias in our target dataset.

2. Background and Related Work

There are so many such biases in languages that it is crucial to identify and measure them, so that the various [3] cultural groupings gain insight into their nature. Researchers have spent decades studying various biases in language, which include how people name objects, places, how they categorize experiences, and in the language used to talk about these topics. Human biases manifest themselves differently, but commonalities can be found. Explicit bias measurements [4] show that the type of prejudices that people are more difficult to acknowledge, such as gender-specific associations, rather than explicit ones. One of the most pervasive and persistent prejudices is that men link with job and sciences and women link with family and liberal art [5]. Additionally, there is an imbalance in the number of words that contain the letters "female" and "male" and have a variety of associations. While there are more terms referring to males, there are many more terms sexualizing females [6].

Gender and racial discrimination were found to be prevalent in many online advertising models, or some of the online models display disparate, discriminatory, representations [7, 8]. A recent study discovered that algorithms used to predict repeat offenders have an unintentional racial bias. [1]. Groups of people use dialects and word choices in social media, for example, age, ethnicity, and location, which use different terms to describe the same topic or situation [2]. The presence of this outcome would imply that specialized data sets and qualified natural language methods are incapable of dealing with minority languages is cited as evidence of this. Biases in learning models for machine learning have been extensively researched [9]. Schmidt [10] discovered the gender disparity in word embeddings and conducted separate debiasing for each gender pair. He demonstrated that there is a significant gender imbalance in terms of gender variants and attempted to eliminate it by stripping out all the words for just the genders. A large agenda effort is dedicated to making consistency expansions (e.g., [11, 12, 13, 14], as far as bias evaluation is concerned. Concepts for modifications to a classification algorithm have been identified in several publications [15, 16, 17] and a recent survey [18] has revealed the approaches available to handle the concept of fairness. The preceding algorithmic fairness studies have largely been in the context of supervised learning. In an unbiased manner, classifications are classified based on whether they used an impartial feature or feature of differing levels of importance, which causes them to bring all who have that feature in a group together in that way. The complexity of word embeddings cannot be restricted to any prior-defined individuals or categories, but they have a known tendency to work better when there

are. As you can see, the above issues demonstrate, however, the same issues appear in the form of overt and indirect bias [19].

Caliskan et al. [20] demonstrated that word embedding techniques such as GloVe reflect human-like semantic biases. The authors demonstrated a high level of bias in pre-trained word vectors. Along with bias against women, they identified bias against people of color. Bolukbasi et al. [21] conducted a thorough examination of gender bias in word embeddings and proposed two debiasing strategies: 'hard debias' and 'soft debias'. The hard debias algorithm accurately determines a word's gender and then refines each vector to be parallel to the gender direction (neutralization). Additionally, they will modify each set of words to achieve parity. By contrast, the soft debias modifies word vectors and compresses the projective function, preserving all inner products while debasing gender-neutral terms. Kaneko and Bollegala [22] proposed a method for deriving four domains from a given vocabulary. For each domain, a set of seed words is used to train an encoder in an autoencoder, with stereotypically gendered words weeded out. Gonen and Goldberg [23] demonstrated that Bolukbasi's approaches that rely on gender direction to define and directly target gender bias end up concealing rather than reducing the bias. Zhou et al. [24] detected the characteristics of gender bias in bilingual word embeddings and gendered grammatical languages (such as Spanish and French). Additionally, they defined semantic and grammatical gender directions to quantify and mitigate gender bias. However, our project is limited to languages with a non-gendered grammar, such as English.

Prior researches are creative overall, but the majority of their works focus researching and resolving how to eliminate bias, which are general in scope. Some of the research examines gender bias, but does not focus on any specific particular field. Additionally, their debias procedure might obliterate some of the original data. On the contrary, our research is devoted to identifying and classifying deviations.

3. Proposed System/Application

3.1 An Overview

Our aim is to determine the bias in our target dataset. First, determine how biased the chosen dataset is by vectorizing the input data through the model and perform calculations. Here is the picture of word-to-vector. Then we use the cos of two vectors to calculate the similarity.

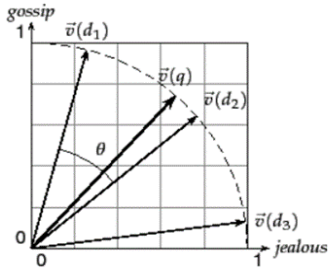


Figure 3.1

Second, determine how to measure bias using math function for data that has been vectorized. We use the cos to calculate the similarity between two vectors.

$$\text{similarity} = \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\sqrt{\sum_{i=1}^{n} B_i^2}},$$

Figure 3.2

Third, we will find some existing experimental methods for bias through the calculation of mathematical models and training of existing data sets. Here is our system flow and use case diagrams.

Finally, implement our own experimental method and perform the above operations to achieve our goal of determining the bias in our target dataset.

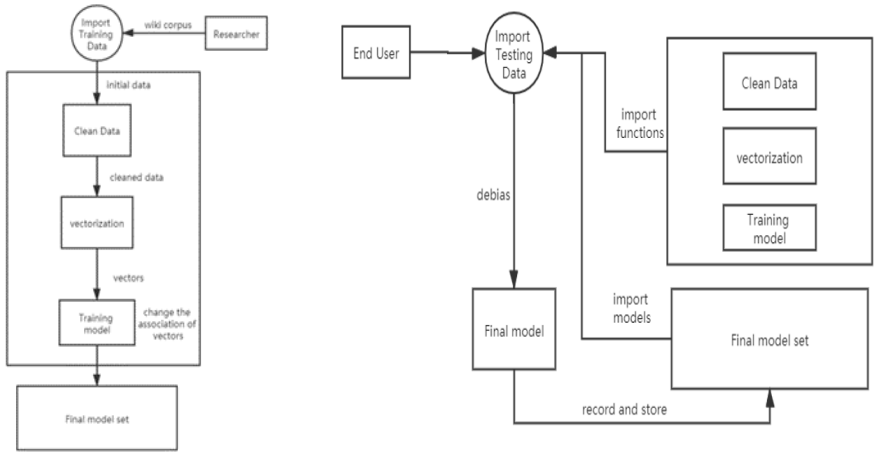


Figure 3.3

3.2 System Requirements

Our key idea is that in the first step we need to find a very suitable data set, and secondly, we need to use the existing function to vectorize our data set, and then process our vectorized data set accordingly to get what we want. The following are the specific steps of our experiment

Functional Requirements

We conducted error analysis for sample problems and tried to find common error categories and classify them. In order to avoid human subjectivity and improve accuracy, we need to use a domain-specific language to quantitatively describe different instances. The analysis of the frequency of errors should be performed on the entire data set, which needs to include true positives. Propose and verify related conjectures, use rules such as rewriting, and verify the conjectures with counterfactual analysis tests. implement our own experimental method and perform the above operations to achieve our goal of determining the bias in our target dataset. AS an End Users. It could identify bias in text messages. Users could use it as a simple auxiliary tool to process some text information, such as text search, machine translation, customer service automatic reply, etc. After find bias, users can identify information when they want to search the data that they interest in. By determine bias, we can create bots that are truly interactive. If we are able to eliminate bias consumers of NLP algorithms will be able to more readily rely on those algorithms to relay their meaning

Our algorithm aims to create the conditions for a truly interactive NLP algorithm. Also, we aim to increase representative accuracy. We create a more welcoming environment. Algorithms may interact with more freedom in a real-world setting

Operating Requirements

This section will discuss the operating requirements that our team is using to develop reducing bias of natural language processing

Programming Languages is Python

• TensorFlow

• Keras

Integrated Development Environment

• VSCode

• Jupiter Notebook

Datasets

•Wikipedia Corpus

•Various textbooks and scholarly articles

Limitations Constraints

Our program could only deal with one type of language at a time. We couldn't determine the voice message and Photo Message we could only process text messages. Also, it would cause errors when converting the text in the image to text format. Another problem is about the Grammar. It cannot deal with semantic ambiguity caused by grammatical problems inherent in the original text. Sentences are often not just literal meanings. Sometimes the answers to some questions require unique answers based on the context.

3.3 Technical Design

The purpose of this report is to showcase two important diagrams of our determined gender bias in the natural language processing system. The first diagram is the use case diagram, which shows how the end users utilize the dataset and models that researchers design and trained before. The second is the system/data flow diagram. It will show how data will process and models will be created through the natural language processing system

Determining bias in natural language processing systems uses a case diagram that begins with the researcher's training data. Researchers are responsible for cleaning, vectorizing, and training data to obtain mature models for detecting bias. End users would import the models to achieve further functions and research.

The system and data flow diagram begins in two ways. Data flow first begins with researchers. Ideally, the first way will be the researchers inputting their training data, such as Wikipedia corpus. The first key process is to clean the data. The second key process is to vectorize our data. And the last key process is to change the association of vectors and obtain our final models. The second way might begin with end-users as they could use or import our functions to their programs.

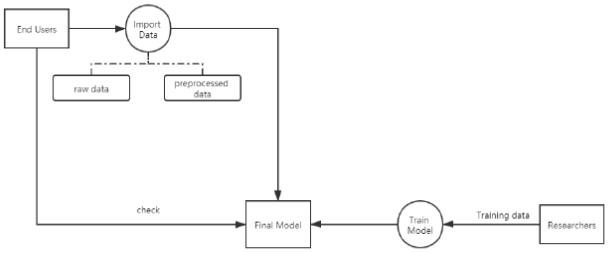


Figure 3.4

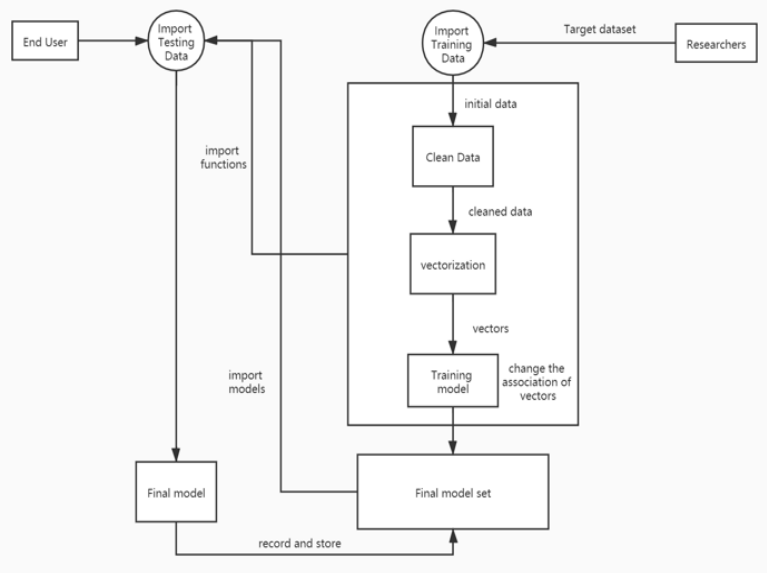


Figure 3.5

3.4 System Implementation

The operating requirements in our team is to use to detect bias of natural language processing. The programming language is Python. We use Python under TensorFlow and Keras environment. Our Integrated Development Environment are VSCode and Jupiter Notebook. We also use Wikipedia Corpus as our data and lookup for Various textbooks and scholarly articles

For this step, we will have to choose a dataset, or multiple datasets, to train and test our AI algorithm on. We use wiki corpus as our training data. This corpus contains the full text of Wikipedia, and it contains 1.9 billion words in more than 4.4 million articles. Most importantly, you can create and use virtual corpora from any of the 4,400,000 articles in the corpus.

Second step. Text vectorization, also known as the "word vector model", is to represent text as a computer-recognizable real number vector. According to the different granularity, the text feature representation can be divided into several levels of words, words, sentences, or chapters. The method of text vectorization is mainly divided into discrete representation and distributed representation. Word2Vec we choose is one of the discrete representations. Word2vec vectorizes all words so that words can be measured quantitatively. Word2vec is based on word embedding and uses the idea of deep learning to predict the words that appear in the context. The word vector after Word2vec training can measure the similarity between words and project all words to k-dimensional vector space. Each word can be represented by a k-dimensional vector, which simplifies the processing of text content to vector operations in a K-dimensional vector space. By reading a lot of words, it can learn and remember words that tend to appear in similar contexts. After sufficient data training, it will generate a 300-dimensional vector for each word in the vocabulary to record words with similar semantics.

The last step. For our training data, this is an important part of our model. First, we will select some text from the wiki corpus as our initial data, and then we will select 80% of them as samples for training, and then 20% of them will be used as test data to fit the results of our data training and Test Results. Finally, our training model is reached. We will attempt to use a Part-of-Speech tagger to break down sentences into their grammatical and relational components using a library called SpacY. From there, we will analyze, for example, the frequency of gendered pronouns with nouns and verbs to help understand how gendered a dataset is, what the most gendered words in the dataset are, and whether there are commonalities between the most gendered words across multiple datasets. By learning and locating debias in natural language processing, future researchers could reduce the workload of locating debias. Our research could help researchers to find out the biased words more comprehensively and quicker, and accomplish their corresponding debiasing work.

3.5 Use of Computer Science Theory and Software Development Fundamentals

In our project, we mainly base on mathematical theoretical knowledge and transform it into computer code. We use the theoretical knowledge of machine learning, which covers the knowledge of probability theory, statistics and complex algorithms, and put the text embedding algorithm and Cosine similarity theory into practice. Word embedding is a type of word representation. It is a technique in which a single word is represented as a real number vector in a predefined vector space, and each word is mapped to a vector. Words with similar meanings have similar representations. It is a general term for the method of mapping words to real number vectors. Cosine similarity measures the similarity between two vectors by measuring the cosine of the angle between them. The cosine value of an angle of 0 degrees is 1, and the cosine value of any other angle is not greater than 1. And its minimum value is negative 1. The cosine of the angle between the two vectors thus determines whether the two vectors are roughly pointing in the same direction. In addition, our research focuses on theoretical knowledge, and our code cannot be used as an application, only for natural language Processed package. Therefore, our software fundamentals the method of testing specific target domains as a reusable coding practice.

4. Experimental Design and Testing

Our aim is to reduce bias present in Natural Language Processing word vectors.

Experimental Setup

- 1. We had to first find a dataset and load that dataset in. This dataset needed to be an already trained word embedding.
- 2. We then had to determine how to measure the similarity between words, and get a measure of how biased words in our dataset are.
- 3. Finally, we looked at target domains using our measure, finding interesting domains for further exploration and study.
- 4. For each experiment we used the Python programming language, and jupyter notebook as aan IDE. We used numpy and scipy for analysis and the gensim library to find our dataset. We also used Tensorflow for modeling.
- 5. The dataset was obtained from the gensim library, and we looked at both a glove embedding and a gensim embedding.

Experiment 1

- 1. We will use cosine similarity and the dot product to demonstrate how biased the embeddings are.
- 2. The previous work (Double-Hard Debias) is already in use in the dataset, so any work we do will improve on that result
- 3. To substantiate our claim, we will again use cosine similarity and/or the dot product and compare those results with the uncleaned dataset
- 4. What was unexpected was that the Double Hard Debias would already be implemented, further complicating the expected, which would be the difficulty of the problem.

Experiment 2

- 1. Following loading our new dataset into python, we did some routine cleaning and analyzed the cosine similarity in the embedded word vectors.

- 2. This time the vectors were trained using the GloVe method, and did have the fidelity to capture details about how frequently different words were used together
- 3. It seemed to have already undergone debiasing during embedding however, as we are not finding the level of bias that we anticipated.
- 4. We are performing extensive statistical analysis to find places where bias exists and try to get an idea of potential debiasing methods.

Results and Analysis

The results were promising. We were able to isolate numerous domains that contain bias. Examples include the domain of finance and the domain of housework. Each domain demonstrated strong bias.

Figure 4.1 is the result of our model calculation. After our vectorization and similarity calculation, we get the similarity of the same words in different fields. we have obtained sevral sets of similarity analysis data. Through the data, we can know that the bias of embedded words already exists, but in the previous work, Double-Hard Debias has already as an existing method used in the data set, we then Used the same method to compare with the uncleaned data set. In figure 4.2 4.3 4.4, this shows our bias for different words in different fields. In Figures 4.5, 4.6, and 4.7, we plot the gender biases in different fields from the results we got. In these three figures, the orange is a measure of how asociated spacewords, our words that are taken from the domain of space travel, with the female gnder identifier, the blue is the male.

Our system will be superior to other similar systems because most hard debias methods are limited and gender bias could still be recovered from embedded geometry. Since word frequency will distort the gender direction, we will use some mathematics methods to eliminate the negative effects of word frequency. The key idea is to generally project the word embedding into the intermediate subspace before using ordinary debias methods, and then reduce gender bias by converting the embedding space into a genderless space.

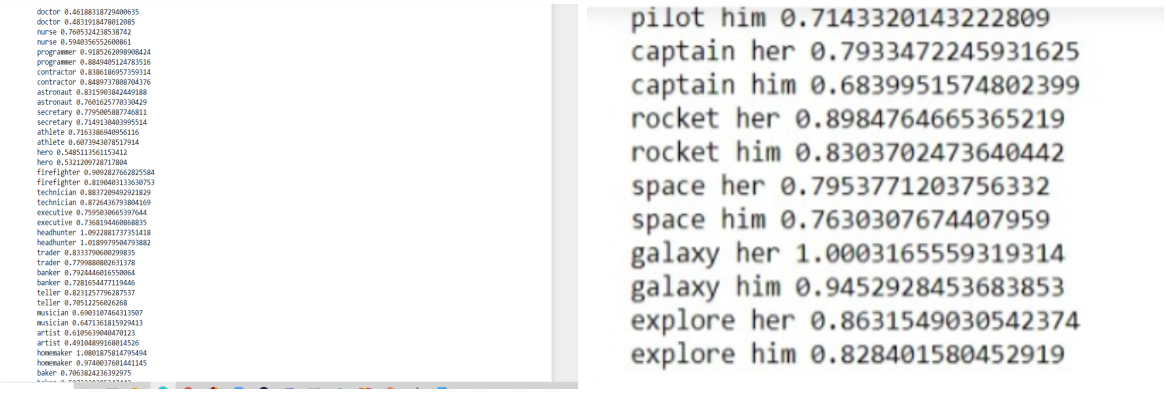


Figure 4.1

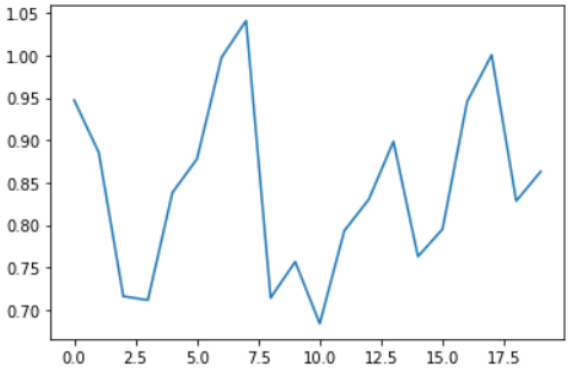


Figure 4.2

Limitations

We are aware that our program is limited in scope. The research on gender bias in NLP is still relatively new, so there is a lack of unified evaluation indicators and benchmarks. At the same time, we also recognize that different applications may require different metrics, and they need adjustments between different concepts of deviation. However, our program is just a determine bias model. Through existing research and experiments, we have noticed that certain detect techniques may introduce noise into the NLP model, resulting in performance degradation. Moreover, the code we write may contain unintentionally implicit biases.As our project is being designed and developed, there would be a few limitations within the system. Our system only could process biases of one type of a language at a time and we focus on English for our project currently. Since we use Wikipedia corpus as our input data/training data, so the system is designed to process text type messages of natural language. We could process voice or photo messages only if they are already converted to text. Therefore, our input is limited to text data and we do not allow other types of input.Training

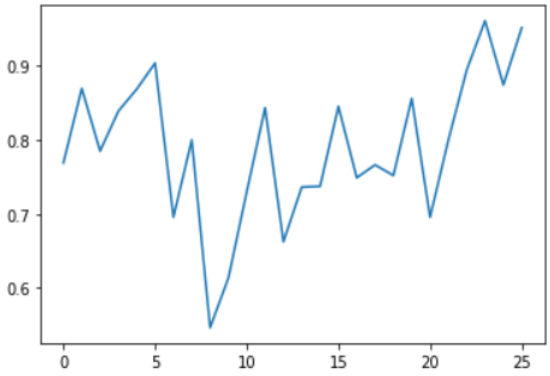


Figure 4.3

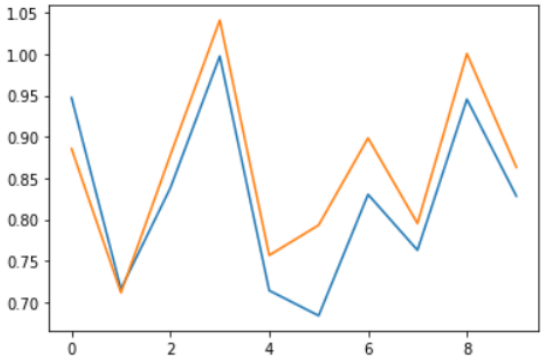


Figure 4.4

word embeddings is a costly and complicated process, so we will be forced to use pre-trained embeddings, which will limit us to word embeddings that may not be as reflective of the overall natural language as they may otherwise be if we could train our own embeddings (e.g., A manual on auto repair will have a different vocabulary than a Shakespeare novel).

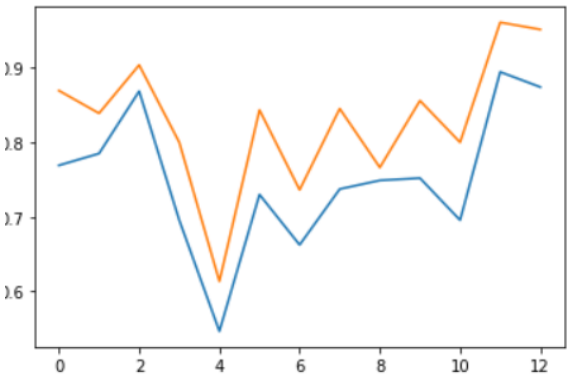


Figure 4.5

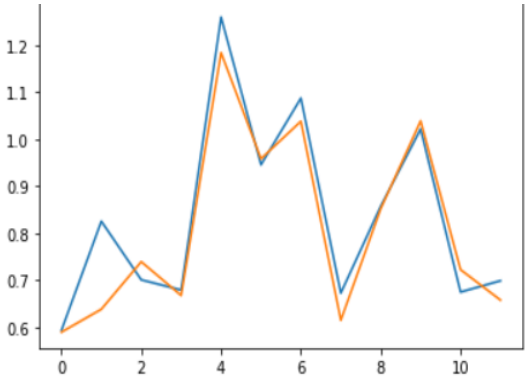


Figure 4.6

5. Ethics and Legal Practices

This project was entirely about ethics. Reducing bias is an important step toward equality and developing truly representative artificial intelligence algorithms. With recent examples like Google's DeepMind revealing the harm that may be done by ignoring this problem, we contend that debiasing measures must keep pace with advancing algorithmic models. Ignoring the debiasing process is highly irresponsible and doubtless unethical. Users should be made aware that such steps were taken and should use any technology developed in full awareness. However, such a process can not be allowed to become a political battleground. The issue of bias, and the issue of ethics in artificial intelligence, have become hot-button issues recently, and this sits at the nexus of both.

6. Effort Sharing

For our team, we do not have a clear division of tasks. At the beginning of the project, all team members participated in the discussion and gave their own ideas, thus determining the goals of our team. During the development stage, all team members participated in the discussion and development of each part, including finding suitable data sets, using corresponding mathematical tools, and reading papers suggested by the project sponsor. Randall and Jiangwei are mainly responsible for finding data sets and algorithms for eliminating bias. Mingwei and Bowen are responsible for processing text information. All team members have participated in algorithm research and make ppt for presentation, and report. If it must quantify our contribution, I can only say that everyone has done 25% (as shown in Table 6.1).

Table 6.1: Effort sharing

Team size	Joint efforts	Randall	Bowen	Jinagwei	Mingwei
4	J (20%)	I (20%)	I (20%)	I (20%)	I (20%)

J = description of tasks jointly performed
I = description of tasks individually performed

7. Conclusion and Future Work

In conclusion, in our project, we learned a lot about machine learning and have a basic understanding of natural language processing. When we developed this project, because our project focused more on studying the deviations in the text, the focus of our project was not on the code, but more on the learning of mathematical theory and the learning of algorithms. At the beginning of the project, we encountered great difficulties, because most of the team members did not have a background in machine learning and natural language processing, and our sponsor helped us a lot. In the future, we will continue to do our project during the summer vacation, because dealing with deviations in the text has always been a difficult part of natural language processing. All our team members will not give up their research in the field of machine learning. We will also use this project as a simple attempt to enter the field of data science.

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